be recorded for every temperature point:

(a) Thermocouple reading
(b) Resistivity; magnetic field off
(c) Resistivity; magnetic field on
(d) Hall voltage, with field forward, off, reversed
(e) Thermocouple reading

Less experienced persons are better advised first to make a measurement of the resistivity only, over the whole temperature range from 80° K to 330° K, and then to measure the Hall voltage separately; as usual, it is advisable to plot the data as it is obtained so as to know where a greater density of measurements is desirable.

3.5 Analysis of Data

Data on Hall effect and resistivity, obtained by students† using a low impurity germanium crystal, are presented and analyzed below.

Table 3.2 gives the raw data; the dimensions of the crystal were $l = 1.09$, $t = 0.183$, $w = 0.143$ cm (see Fig. 3.20). A fixed voltage of 1.32 V was applied across the long end of the sample, and the magnetic field was (800 ± 40) gauss. The Hall voltage was measured across the t dimension of the crystal.

From the data of the Table 3.2 the following plots were made:

(a) $\rho = 1/\sigma = RA/l$, hence‡ $\rho = R \times 2.42 \times 10^{-2}$ ohms-cm; this is shown in a semilog plot against $1/T$ in Fig. 3.25. We note that for $T < 290°$ K, conduction is due mainly to the impurity carriers: this is the *extrinsic region*. For $T > 280°$ K, electrons are transferred copiously from the valence band into the conduction band and the crystal is in the *intrinsic region*. From the slope of the intrinsic region and making use of Eq. 3.3b, we have $\rho \propto 1/n \propto \exp(E_g/2kT)$ and thus $\ln \rho = E_g/2kT$. Hence

$$\frac{E_g}{2k} = \frac{\Delta \log_{10}}{\Delta(1/T)} \frac{1}{\log_{10} e} = \frac{1.81 \times 10^3}{0.4343} \tag{3.16}$$

which leads to $E_g = 0.72 \pm 0.07$ eV, in agreement with the accepted value.

(b) A log-log plot of the resistivity in the extrinsic region against $1/T$ shown in Fig. 3.26. If a power law as in Eq. 3.7 is applicable, we would have $\rho \propto (1/T)^\alpha$ and hence

$$\alpha = \frac{\Delta \log \rho}{\Delta \log (1/T)} = -2.0 \pm 0.1 \tag{3.17}$$

† E. Yadlowski and P. Schreiber, class of 1962.
‡ Note that here R is the resistance of the sample and not the Hall coefficient R_H.

TABLE 3.2
Data on Hall Effect and Resistivity Measurements of Germanium

V_H (Volts) Hall voltage	R_m (Ohms) Magneto-resistance	R (Ohms) Resistance	T_{mv} (millivolts) Thermocouple	T (°K)
0.05 ± 0.002	157.8 ± 0.2	142.0 ± 0.2	5.37	84
0.047 ± 0.002	172.0 ± 0.2	152.0 ± 0.2	5.28	86
0.041 ± 0.002	183.9 ± 0.2	168.0 ± 0.2	5.20	93
0.038 ± 0.002	200.0 ± 0.2	184.9 ± 0.2	5.10	100
0.038 ± 0.002	207.0 ± 0.1	192.0 ± 0.2	5.036	103
0.0375 ± 0.002	219.0 ± 0.2	202.2 ± 0.1	5.00	105
0.0360 ± 0.002	229.0 ± 0.2	215.0 ± 0.4	4.94	111
0.0340 ± 0.002	238.0 ± 0.2	226.0 ± 0.4	4.86	114
0.0330 ± 0.002	261.0 ± 0.2	247.0 ± 0.1	4.78	121
0.0300 ± 0.002	290.0 ± 0.2	276.0 ± 0.2	4.65	127
0.0280 ± 0.002	321.0 ± 0.2	306.0 ± 0.5	4.53	131
0.0275 ± 0.002	335.0 ± 0.5	321.0 ± 0.5	4.45	137
0.0230 ± 0.002	371.0 ± 0.5	360.0 ± 0.5	4.30	143
0.0230 ± 0.002	407.0 ± 0.5	396.0 ± 1	4.15	150
0.0208 ± 0.002	457.5 ± 0.5	446.0 ± 1	3.95	155
0.0204 ± 0.002	495.0 ± 0.5	482.5 ± 2	3.84	164
0.0185 ± 0.001	559.0 ± 0.5	550.0 ± 1	3.60	172
0.0180 ± 0.001	610.0 ± 0.5	595.0 ± 2	3.40	182
0.0145 ± 0.0005	702.0 ± 1	700.0 ± 1	3.10	193
0.0150 ± 0.0005	827.0 ± 2	820.0 ± 1	2.72	199
0.0144 ± 0.0003	855.0 ± 1	850.0 ± 1	2.60	203
0.0131 ± 0.0003	888.0 ± 1	875.0 ± 1	2.45	212
0.0115 ± 0.0003	955.0 ± 1	950.0 ± 1	2.20	241
0.0095 ± 0.0001	1270.0 ± 5	1270.0 ± 5	1.20	253
0.0075 ± 0.0002	1440.0 ± 3	1440.0 ± 3	0.69	262
0.0059 ± 0.0002	1500.0 ± 3	1500.0 ± 3	0.45	260
0.0065 ± 0.0002	1460.0 ± 3	1450.0 ± 5	0.65	263
0.0075 ± 0.0002	1500.0 ± 3	1500.0 ± 4	0.380	266
0.0075 ± 0.0002	1570.0 ± 3	1560.0 ± 6	0.180	270
0.0067 ± 0.0002	1650.0 ± 3	1645.0 ± 5	0.000	273
0.0061 ± 0.0002	1700.0 ± 5	1700.0 ± 5	−0.400	282
0.0055 ± 0.0002	1660.0 ± 5	1665.0 ± 5	−0.630	286
0.0050 ± 0.0002	1610.0 ± 5	1590.0 ± 5	−0.860	290
0.0038 ± 0.0001	1455.0 ± 5	1440.0 ± 5	−1.10	296
0.0032 ± 0.0001	1340.0 ± 5	1310.0 ± 5	−1.22	302
0.0018 ± 0.0001	980.0 ± 5	950.0 ± 5	−1.44	305
0.0008 ± 0.0001	795.0 ± 3	785.0 ± 2	−1.67	312
0.00035 ± 0.00005	654.0 ± 2	632.0 ± 1	−1.86	315
0.00001 ± 0.00002	566.0 ± 2	545.0 ± 1	−2.00	318
0.00000 ± 0.000050	505.0 ± 1	488.0 ± 2	−2.15	323
−0.0002 ± 0.000050	395.0 ± 2	390.0 ± 2	−2.30	326
−0.0006 ± 0.00005	303.0 ± 2	293.0 ± 2	−2.65	329
−0.0008 ± 0.00005	225.0 ± 2	225.0 ± 2	−3.05	337
−0.0010 ± 0.0001	178.0 ± 1	178.0 ± 1	−3.33	346

Since in this region the carrier density is constant, this gives for the mobility a dependence

$$\mu = CT^{-2.6} \tag{3.18}$$

which is in disagreement with the prediction of Eq. 3.7. It is, however, the correct value for germanium, indicating that the simplified calculations used in deriving Eq. 3.7 are not completely adequate.

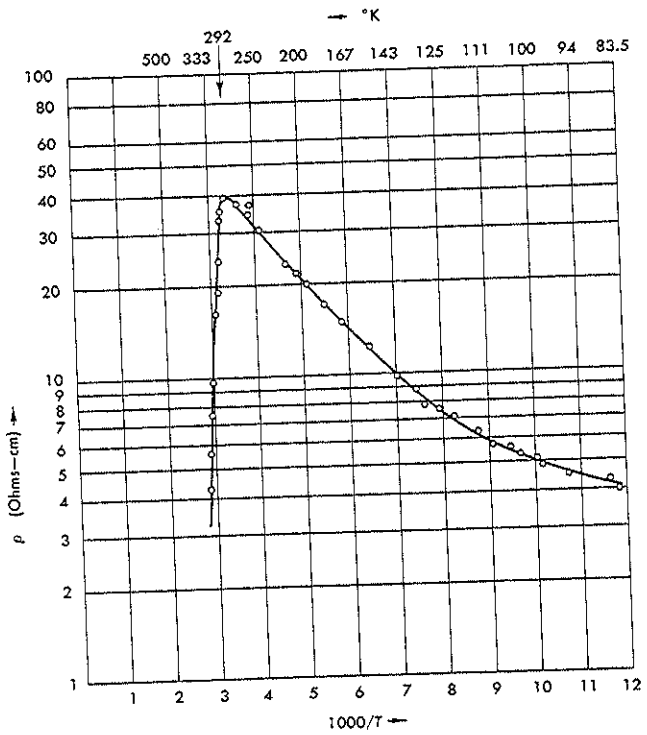


FIG. 3.25 The resistivity of a pure germanium crystal as a function of inverse temperature. For $T < 290°$ K, conduction is due mainly to the impurity carriers (extrinsic region); for $T > 290°$ K, conduction is due to electrons transferred to the conduction band (and the corresponding holes created in the valence band): this is the intrinsic region.

resistivity ρ vs 1/Temp

Turning now to the Hall-voltage measurements of Table 3.2, we can form the quantities defined by Eq. 3.8c.

$$\frac{V_H}{V_S} = \phi = \mu H \frac{t}{l}$$

hence

$$R_H = \phi R \frac{w}{H}$$

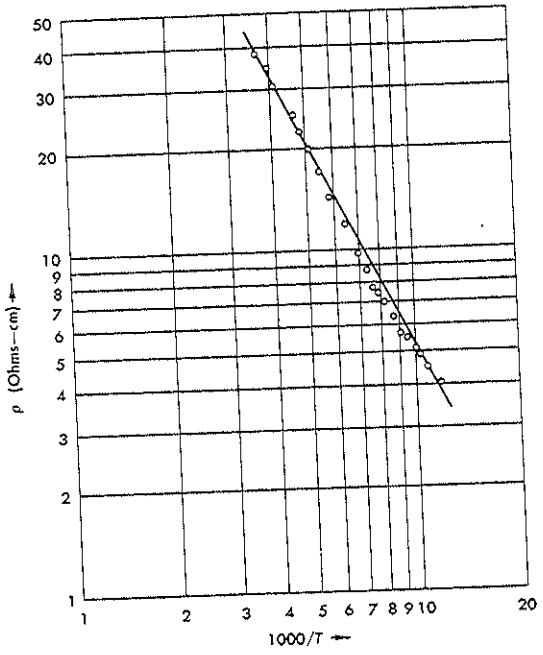


FIG. 3.26 A log-log plot of the resistivity of germanium in the extrinsic region versus $1/T$. It is assumed that the number of carriers is independent of T since saturation of the impurity carriers has already been reached.

log of resistivity vs. 1/temp in the extrinsic region

and the Hall mobility

$$\mu_H = R_H\sigma = \phi\left(\frac{l}{l}\right)H^{-1}$$

The Hall mobility so obtained is shown in a log-log plot against T in Fig. 3.27. Since the Hall coefficient changes sign, we can immediately recognize that the crystal is of the p type; the inversion temperature of this particular sample is found to be

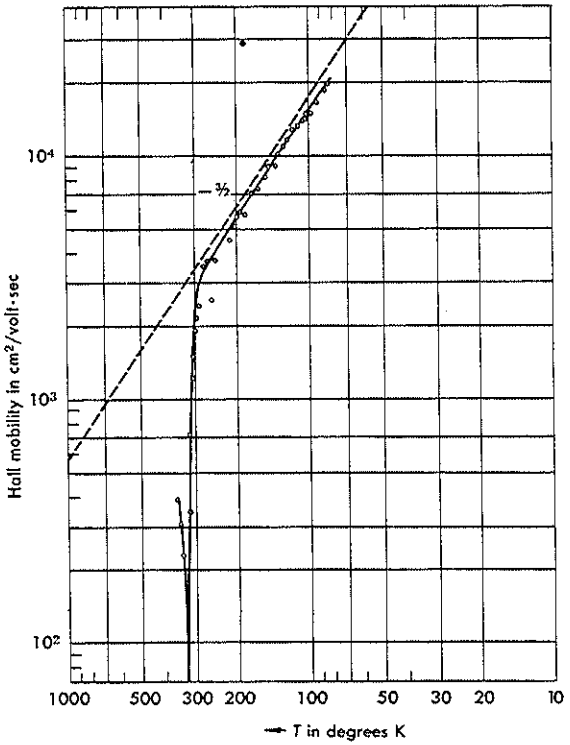


FIG. 3.27 Log-log plot of the "Hall-mobility" (Hall coefficient × conductivity) versus T. We note that the Hall coefficient becomes zero at $T = 323°$ K (inversion temperature) and changes sign beyond that point. Since negative values cannot be shown on the log plot for $T > 323°$ K, the Hall mobility of the reverse sign is again shown in the same graph. Note also the $T^{-3/2}$ dependence of the Hall mobility in the extrinsic region.

Hall Mobility vs. Temp.

$$T_0 = (323 \pm 3)° \text{ K}$$

From the slope of the μ_H curve in the extrinsic region in Fig. 3.27, we obtain

$$\mu_H = CT^{-3/2} \tag{3.19}$$

which is different from Eq. 3.18 and is in agreement with conclusions of other observers; this is the reason why a distinction between the Hall mobility μ_H and the drift mobilities μ_D obtained from resistivity measurements is made.

By extrapolating Eq. 3.19 to the inversion temperature, we obtain the hole mobility† at $T = T_0 = 323°$;

$$\mu_H(h) \doteq 2.7 \times 10^3 \text{ cm}^2/\text{V-sec.} \tag{3.20a}$$

We can now apply the analysis indicated in Section 3.4, which led to Eq. 3.11. From Fig. 3.25 we have $R_e(T = T_0) = 2150$ ohms and $R_0 = 500$ ohms leading to $b = 1.31 \pm 0.2$. Thus we obtain for the electron mobility at $T = T_0 = 323°$

$$\mu_H(e) = 3.5 \times 10^3 \text{ cm}^2/\text{V-sec,} \tag{3.20b}$$

both results being in agreement with the accepted values.

From the Hall coefficient in the extrinsic region, we can also obtain an order of magnitude for the density of impurity carriers. Since in that region only one type of carrier is present, $ne = 1/R_H$, and since

$$R_H = \frac{\phi Rw}{H} \simeq 8 \times 10^5 \text{ cm}^3/\text{coulomb}$$

$$n \simeq 8 \times 10^{12}/\text{cm}^3 \tag{3.21}$$

which is reasonable for this sample, indicating an impurity concentration of the order of two parts in 10^{10}.

From the data of Table 3.2 it can be further noticed that the resistance of the sample changes when the magnetic field is turned on. This phenomenon, called magnetoresistance, is due to the fact that the drift velocity of all carriers is not the same. With the magnetic field on, the Hall voltage $V = E_y l = |\mathbf{v} \times \mathbf{H}|$ compensates exactly the Lorentz force for carriers with the *average* velocity; slower carriers will be overcompensated, and faster ones undercompensated, resulting in trajectories that are not along the applied external field. This results in an effective decrease of the mean free path and hence an increase in resistivity.

† Note that 1 gauss = 10^{-4} weber/m² = 10^{-8} V-sec/cm².